Name; inshal nasir

Section : A

Roll no : 38

```
@23-22412-038-sys →/workspaces/-inshal-nasir-38- (main) $ aws ia
m create-group --group-name SoftwareEngineering
{
    "Group": {
        "Path": "/",
        "GroupName": "SoftwareEngineering",
        "GroupId": "AGPAZCN5ZS5DAB4XH7OPZ",
        "Arn": "arn:aws:iam::623705167686:group/SoftwareEngineeri
ng",
```

Q1.create group

```
@23-22412-038-sys →/workspaces/-inshal-nasir-38- (main) $ aws ia
m get-group --group-name SoftwareEngineering
{
    "Users": [],
    "Group": {
        "Path": "/",
        "GroupName": "SoftwareEngineering",
        "GroupId": "AGPAZCN5ZS5DAB4XH7OPZ",
        "Arn": "arn:aws:iam::623705167686:group/SoftwareEngineeri
ng",
        "CreateDate": "2026-01-19T07:41:35+00:00"
    }
}
```

q1_group_details

```
@23-22412-038-sys →/workspaces/-inshal-nasir-38- (main) $ aws ia
m create-user --user-name Inshal
{
    "User": {
        "Path": "/",
        "UserName": "Inshal",
        "UserId": "AIDAZCN5ZS5DGDMQX5D6O",
        "Arn": "arn:aws:iam::623705167686:user/Inshal",
        "CreateDate": "2026-01-19T07:45:21+00:00"
    }
}
```

q1-create-user

```
@23-22412-038-sys →/workspaces/-inshal-nasir-38- (main) $ aws iam get-user
 --user-name Inshal
{
    "User": {
        "Path": "/",
        "UserName": "Inshal",
        "UserId": "AIDAZCN5ZS5DGDMQX5D6O",
        "Arn": "arn:aws:iam::623705167686:user/Inshal",
        "CreateDate": "2026-01-19T07:45:21+00:00"
    }
}
```

q1_user_details

```
@23-22412-038-sys →/workspaces/-inshal-nasir-38- (main) $ aws iam add-user
-to-group --user-name Inshal --group-name SoftwareEngineering
```

Q1_add_user_to_group

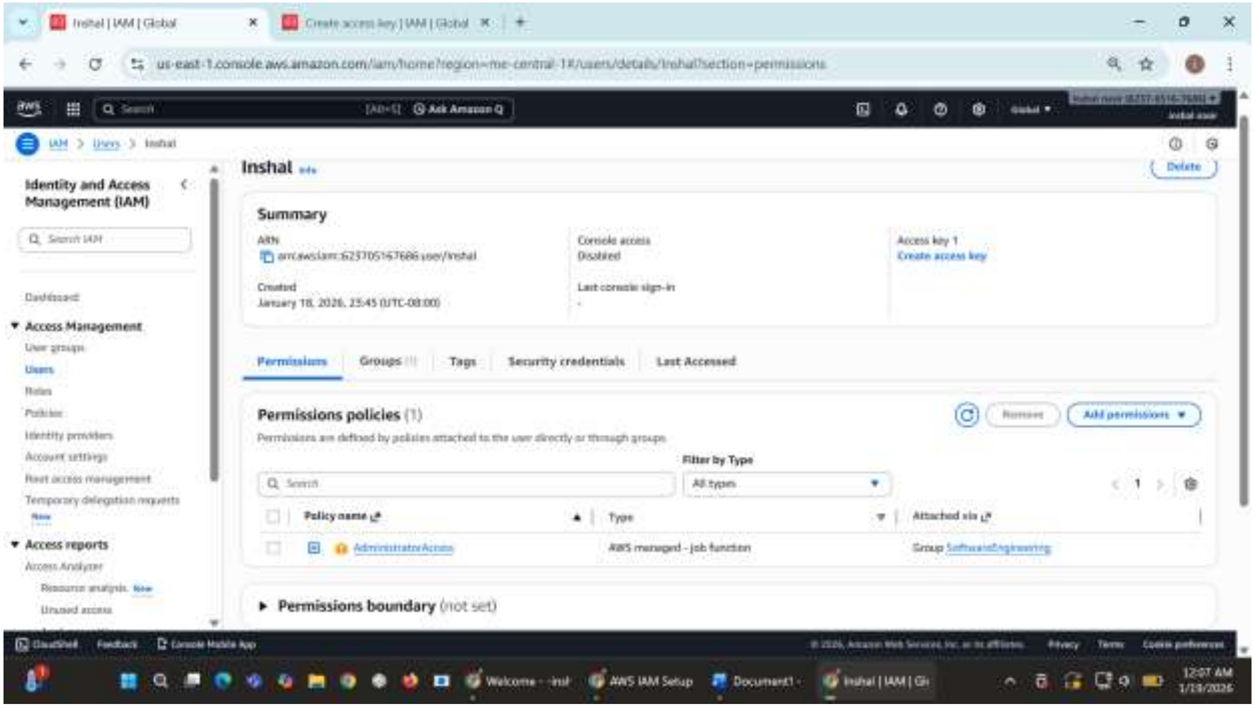
```
@23-22412-038-sys →/workspaces/-inshal-nasir-38- (main) $ aws iam get-grou
p --group-name SoftwareEngineering
{
    "Users": [
        {
            "Path": "/",
            "UserName": "Inshal",
            "UserId": "AIDAZCN5ZS5DGDMQX5D6O",
            "Arn": "arn:aws:iam::623705167686:user/Inshal",
            "CreateDate": "2026-01-19T07:45:21+00:00"
        }
    ],
    "Group": {
        "Path": "/",
        "GroupName": "SoftwareEngineering",
        "GroupId": "AGPAZCN5ZS5DAB4XH7OPZ",
        "Arn": "arn:aws:iam::623705167686:group/SoftwareEngineering",
        "CreateDate": "2026-01-19T07:41:35+00:00"
    }
}
```

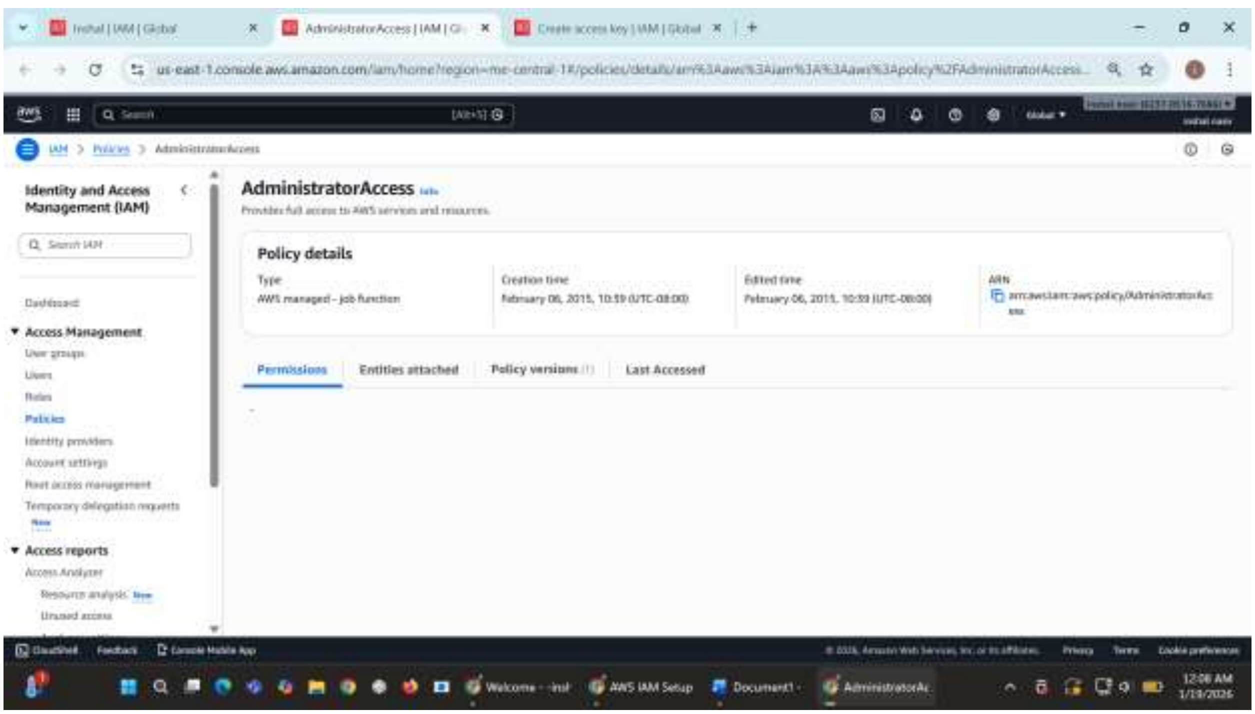
q1_group_membership.png

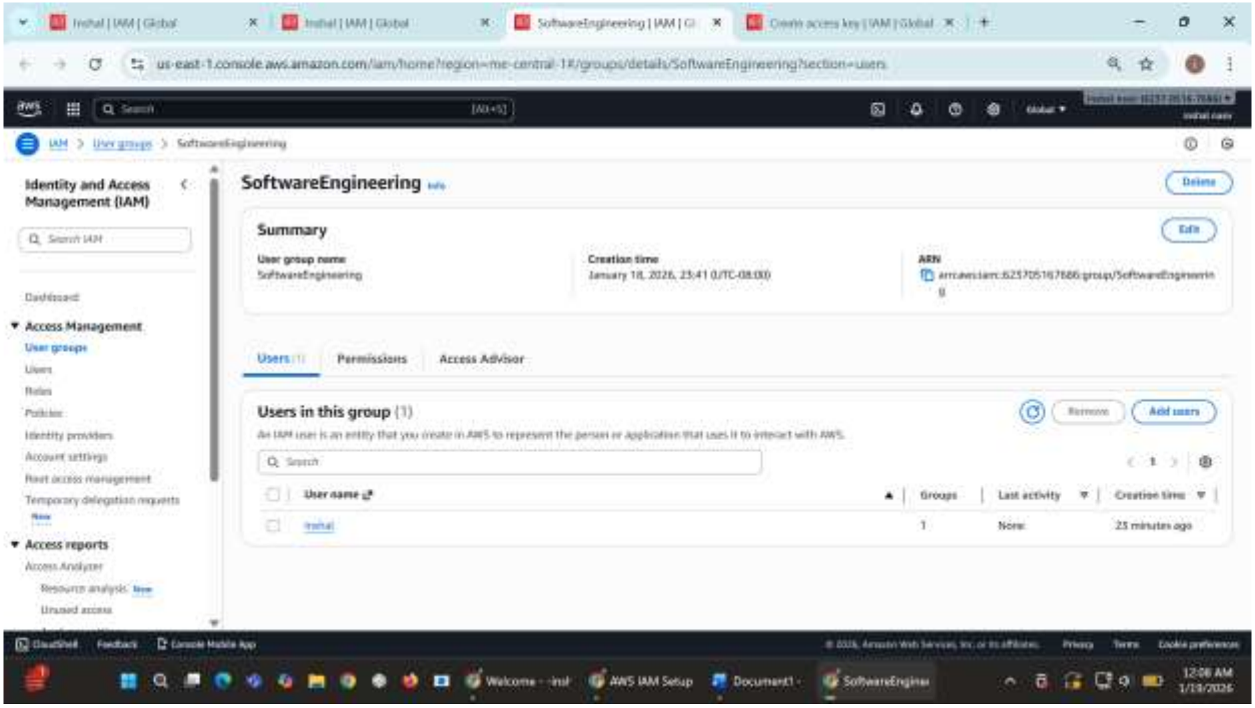
```
@23-22412-038-sys →/workspaces/-inshal-nasir-38- (main) $ aws iam list-pol
icies --scope AWS --query "Policies[?PolicyName=='AdministratorAccess']"
[
    {
        "PolicyName": "AdministratorAccess",
        "PolicyId": "ANPAIWMBCKSKIEE64ZLYK",
        "Arn": "arn:aws:iam::aws:policy/AdministratorAccess",
        "Path": "/",
        "DefaultVersionId": "v1",
        "AttachmentCount": 1,
        "PermissionsBoundaryUsageCount": 0,
        "IsAttachable": true,
        "CreateDate": "2015-02-06T18:39:46+00:00",
        "UpdateDate": "2015-02-06T18:39:46+00:00"
    }
]
```

```
@23-22412-038-sys →/workspaces/-inshal-nasir-38- (main) $ aws iam attach-g
roup-policy --group-name SoftwareEngineering --policy-arn arn:aws:iam::aws:
policy/AdministratorAccess
@23-22412-038-sys →/workspaces/-inshal-nasir-38- (main) $
```

```
@23-22412-038-sys →/workspaces/-inshal-nasir-38- (main) $ aws iam list-att
ached-group-policies --group-name SoftwareEngineering
{
    "AttachedPolicies": [
        {
            "PolicyName": "AdministratorAccess",
            "PolicyArn": "arn:aws:iam::aws:policy/AdministratorAccess"
        }
    ]
}
@23-22412-038-sys →/workspaces/-inshal-nasir-38- (main) $
```


Inshal
Delete
Identity and Access Management (IAM)
Summary
ARN
arn:aws:iam::623705167686:user/Inshal
Console access
Disabled
Access key 1
Create access key
Created
January 18, 2026, 23:45 (UTC-08:00)
Last console sign-in
Access Management
Access reports
Permissions
Groups
Tags
Security credentials
Last Accessed
Permissions policies (1)
Add permissions
Filter by Type
All types
Policy name
Type
Attached via
AdministratorAccess
AWS managed - job function
Group SoftwareEngineering
Permissions boundary (not set)
AWS IAM Setup


AdministratorAccess
Identity and Access Management (IAM)
Policy details
Type
AWS managed - job function
Creation time
February 06, 2015, 10:39 (UTC-08:00)
Edited time
February 06, 2015, 10:39 (UTC-08:00)
ARN
Access Management
Policies
Access reports
Permissions
Entities attached
Policy versions
Last Accessed
AWS IAM Setup



Question 2

```
provider "aws" {
  region  = "me-central-1"
  profile = "default"
}
resource "aws_vpc" "myapp_vpc" {
  cidr_block = var.vpc_cidr_block
  tags = {
    Name = "${var.env_prefix}-vpc"
  }
}

resource "aws_subnet" "myapp_subnet" {
  vpc_id            = aws_vpc.myapp_vpc.id
  cidr_block        = var.subnet_cidr_block
  availability_zone = var.availability_zone
  tags = {
    Name = "${var.env_prefix}-subnet-1"
  }
}
```

```
resource "aws_internet_gateway" "igw" {
  vpc_id = aws_vpc.myapp_vpc.id
  tags = {
    Name = "${var.env_prefix}-igw"
  }
}

resource "aws_route_table" "rt" {
  vpc_id = aws_vpc.myapp_vpc.id
  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.igw.id
  }
  tags = {
    Name = "${var.env_prefix}-rt"
  }
}

resource "aws_route_table_association" "rta" {
  subnet_id      = aws_subnet.myapp_subnet.id
  route_table_id = aws_route_table.rt.id
}
```

```
resource "aws_route_table_association" "rta" {
  subnet_id      = aws_subnet.myapp_subnet.id
  route_table_id = aws_route_table.rt.id
}
data "http" "my_ip" {
  url = "https://icanhazip.com"
}

locals {
  my_ip = "${chomp(data.http.my_ip.body)}/32"
}
```

```
resource "aws_security_group" "default_sg" {
  vpc_id = aws_vpc.myapp_vpc.id
  name   = "${var.env_prefix}-default-sg"

  ingress {
    description = "SSH from my IP"
    from_port   = 22
    to_port     = 22
    protocol    = "tcp"
    cidr_blocks = [local.my_ip]
  }

  ingress {
    description = "HTTP from anywhere"
    from_port   = 80
    to_port     = 80
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
    description = "HTTPS from anywhere"
```

```
  resource "aws_key_pair" "serverkey" {
  key_name   = "serverkey"
  public_key = file("/home/codespace/.ssh/id_ed25519.pub")
}
```

```
 GNU nano 7.2                    terrafrom.tfvars *
vpc_cidr_block     = "10.0.0.0/16"
subnet_cidr_block = "10.0.10.0/24"
availability_zone = "me-central-1a"
env_prefix         = "dev"
instance_type      = "t3.micro"
```

```
Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to
ee
any changes that are required for your infrastructure. All Terraform comma
ds
should now work.
```

```
@23-22412-038-sys ➜/workspaces/-inshal-nasir-38-/terraform_lab (main) $ te
rraform apply
data.http.my_ip: Reading...
data.http.my_ip: Read complete after 0s [id=https://icanhazip.com]

Terraform used the selected providers to generate the following execution
plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.web will be created
  + resource "aws_instance" "web" {
      + ami                                  = "ami-0f2b4fc905b0bd1f1"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = true
```

Question 3

```
@23-22412-038-sys ➜/workspaces/-inshal-nasir-38-/terraform_lab (main) $ mk
dir -p Lab_exam/ansible
cd Lab_exam/ansible
@23-22412-038-sys ➜.../-inshal-nasir-38-/terraform lab/Lab exam/ansible (m
```

```
[frontend]
<frontend-public-ip>

[backends]
<backend1-public-ip>
<backend2-public-ip>
<backend3-public-ip>

[all:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/.ssh/id_ed25519
```

```
  GNU nano 7.2                    playbook.yml *
 name: Configure EC2 Web Server
  hosts: ec2
  become: true
  tasks:

    - name: Update all packages
      yum:
        name: "*"
        state: latest

    - name: Stop Nginx if running
      service:
        name: nginx
        state: stopped
        enabled: no
      ignore_errors: yes
```